**MATHEMATICAL ANALYSIS 1**
**HOMEWORK 7**

(1) **Products of expansions.** Let $p_n(x) = a_0 + a_1 x + \cdots + a_n x^n$ and $q_m(x) = b_0 + b_1 x + \cdots + b_m x^m$ be polynomials of degrees $n$ and $m$ (so $a_n \neq 0$, $b_m \neq 0$).

(a) Write the product $p_n(x)\, q_m(x)$ in the form $\sum_k c_k x^k$: what is its degree, and what is the coefficient $c_k$? *(Pay attention: $n$ and $m$ might be different.) Hint: first, as an example, try expanding the product $(a_0+a_1x+a_2x^2)(b_0+b_1x)$ and collect the terms with $x^0$, $x^1$, $x^2$, $x^3$. Which products $a_i b_j$ contribute to the coefficient of $x^k$?*

(b) Suppose that, as $x \to 0$,

$$f(x) = p_n(x) + o(x^n) \qquad \text{and} \qquad g(x) = q_m(x) + o(x^m).$$

Let $N = \min\{n, m\}$. Show that $f(x)\, g(x) = r(x) + o(x^N)$ as $x \to 0$, where $r$ is the polynomial obtained from $p_n\, q_m$ by deleting all the terms of degree larger than $N$. Name the rules (Proposition 8.1, Corollary 8.2) that you use at each step.

(c) If $f(x) = 1 + 2x - x^2 + o(x^2)$ and $g(x) = 3 - x + o(x)$ as $x \to 0$, write $f(x)\, g(x)$ in the same form. Can you determine the coefficient of $x^2$ in $f(x)\, g(x)$? Explain.

(2) **Limits with equivalences.** Compute the following limits, using the fundamental limits written as equivalences (Section 8.1.5) together with Proposition 8.3 and Corollary 8.4:

(a) $\displaystyle\lim_{x\to 0} \frac{\sin(3x)\, \ln(1+2x)}{x\,(e^x - 1)}$

(b) $\displaystyle\lim_{x\to 0} \frac{1-\cos(2x)}{x \ln(1+3x)}$

(c) $\displaystyle\lim_{x\to 0} \frac{\sqrt[3]{1+6x} - 1}{\sin(4x)}$

(d) $\displaystyle\lim_{x\to 0} \frac{e^{x^2} - 1}{1 - \cos x}$

(e) $\displaystyle\lim_{x\to 0} \frac{\ln(1+3x) + x^2}{2x + \sin^2 x}$

(f) $\displaystyle\lim_{x\to +\infty} x^2 \left(1 - \cos\frac{1}{x}\right)$

(g) $\displaystyle\lim_{x\to +\infty} x \ln\left(1 + \frac{2}{x}\right)$

(h) $\displaystyle\lim_{x\to +\infty} \left(1 + \frac{3}{x}\right)^{2x}$

(i) $\displaystyle\lim_{x\to \frac{\pi}{2}^-} \left(\frac{1}{\cos x} - \tan x\right)$

(j) $\displaystyle\lim_{n\to\infty} n\left(e^{1/n} - 1\right)$

(k) $\displaystyle\lim_{n\to\infty} n^2 \left(1 - \cos\frac{\pi}{n}\right)$

*Hint for (i): take $y = \frac{\pi}{2} - x$.*

(3) **Equivalences cannot be used in sums.** A student argues as follows: *since* $\sin(2x) \sim 2x$ *and* $\sin x \sim x$ *as* $x \to 0$,

$$\lim_{x\to 0} \frac{\sin(2x) - 2\sin x}{x^3} = \lim_{x\to 0} \frac{2x - 2x}{x^3} = 0.$$

(a) Explain what is wrong with this argument.

(b) Compute the limit correctly. *Hint:* $\sin(2x) = 2\sin x \cos x$.

(4) **The hierarchy of infinities.**

(a) Using the ordering of infinite functions and sequences (Section 8.2), compute:

(i) $\displaystyle\lim_{x\to+\infty} \frac{x^5 + e^x}{2e^x + x^2}$

(ii) $\displaystyle\lim_{x\to+\infty} \frac{x^3 + 2^x}{3^x + x^{10}}$

(iii) $\displaystyle\lim_{x\to+\infty} \frac{\ln x}{\sqrt{x}}$

(iv) $\displaystyle\lim_{x\to+\infty} \frac{\ln x}{x^2}$

(v) $\displaystyle\lim_{n\to\infty} \frac{n! + 3^n}{n^n + n^{10}}$

(b) *Running times of algorithms.* Four algorithms solve the same problem; on an input of size $n$ they need $2^n$, $n\log_2 n$, $n!$ and $n^3$ elementary operations, respectively. Order them from the fastest to the slowest for large $n$, and justify the order. A computer performs $10^9$ operations per second: how long does each algorithm take for $n = 50$?

(5) **Order and principal part.**

(a) Determine the order and the principal part with respect to $\varphi(x) = \frac{1}{x}$ as $x \to +\infty$ of the function $f(x) = \sin\left(\sqrt{x^2-1} - x\right)$.

(b) As $x \to +\infty$, the function $f(x) = \ln\left(9 + \sin\frac{2}{x}\right) - 2\ln 3$ can be written as $f(x) = \frac{b}{x^\alpha} + o(x^{-\alpha})$. Find $b$ and $\alpha$.

(c) Determine the order and the principal part with respect to $\varphi(x) = x$ as $x \to 0$ of the function $f(x) = \frac{e^x}{1+x^2} - 1$.

(6) **Uniqueness of the order.** Let $f : \mathbb{R} \to \mathbb{R}$ be infinite or infinitesimal at $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$, and let $\varphi$ be the corresponding comparison function of Section 8.3.

(a) State the definition of the *order* $\alpha$ of $f$ at $x_0$ with respect to $\varphi$.

(b) Prove that the order is unique: if $f$ has order $\alpha$ and also order $\beta$ with respect to $\varphi$, then $\alpha = \beta$. *Hint: look at the quotient* $|\varphi|^\alpha / |\varphi|^\beta$.

(7) **More orders and principal parts.**

(a) Determine the order and the principal part of each of the following functions with respect to the indicated $\varphi$:

(i) $\sqrt{1+x^2} - 1$ as $x \to 0$, with respect to $\varphi(x) = x$;

(ii) $1 - \cos\sqrt{x}$ as $x \to 0^+$, with respect to $\varphi(x) = x$;

(iii) $x^3 + \sqrt{x}$ as $x \to 0^+$, and as $x \to +\infty$, with respect to $\varphi(x) = x$;

(iv) $\frac{x^2+1}{x-1}$ as $x \to 1$, with respect to $\varphi(x) = \frac{1}{x-1}$.

(b) *How far is the horizon?* From a height $h > 0$ above the sea, the distance to the horizon is $d(h) = \sqrt{(R+h)^2 - R^2}$, where $R \approx 6371$ km is the radius of the Earth (the line of sight is tangent to the Earth). Show that $d$ is infinitesimal as $h \to 0^+$, and find its order and principal part with respect to $\varphi(h) = h$. Deduce the rule of thumb "$d \approx 3.57\sqrt{h}$ km, with $h$ in meters" and use it for $h = 1.7$ m (standing on the beach) and $h = 100$ m (a lighthouse). How does $d(h)$ behave as $h \to +\infty$?

(8) **Asymptotes.**

(a) Find all the asymptotes (vertical, horizontal and oblique) of the following functions, on their natural domains:

(i) $\dfrac{x^2+3x-1}{x+2}$

(ii) $\sqrt{x^2+4x}$

(iii) $x\,e^{1/x}$

(iv) $x \arctan x$

(v) $e^{\sqrt{x}}$

(vi) $\dfrac{1}{(x^2+4)^2}$

(vii) $\ln(\sin x)$, $x \in (0, \pi)$

(viii) $\sqrt[3]{x^2-3x}$

(ix) $\arctan\dfrac{x}{x-1}$

*Hint for (iv): for* $x > 0$, $\frac{\pi}{2} - \arctan x = \arctan\frac{1}{x}$.

(b) *A falling body with air resistance.* A body of mass $m$ falls from rest; the air exerts on it a resistance force proportional to its velocity, $kv$ with $k > 0$. Physics shows that at time $t \geq 0$ its velocity and the distance it has covered are

$$v(t) = v_T\left(1 - e^{-t/\tau}\right), \qquad s(t) = v_T t - v_T\tau\left(1 - e^{-t/\tau}\right),$$

where $v_T = \frac{mg}{k}$ and $\tau = \frac{m}{k}$.

(i) Show that $\lim_{t\to+\infty} v(t) = v_T$. This is why $v_T$ is called the *terminal velocity*: it is the speed at which the air resistance balances gravity.

(ii) Show that $s$ has an oblique asymptote as $t \to +\infty$ and find it. Interpret the result: compared with a body that moves with constant velocity $v_T$ from the start, how far behind is the falling body after a long time?

(9) **Zeros on an open interval.** In this problem we prove Corollary 9.2 of the lecture notes (the existence of a zero of a continuous function on an open interval $(\alpha, \beta)$, where $\alpha$ may be $-\infty$ and $\beta$ may be $+\infty$). You are guided in the steps below.

(a) State Corollary 9.2.

(b) Suppose that $\ell_\alpha < 0 < \ell_\beta$ (state that the case $\ell_\alpha > 0 > \ell_\beta$ is similar, and is therefore omitted). Fix a point $c \in (\alpha, \beta)$.

(c) Using Theorem 6.2 (local sign, applied to the right limit at $\alpha$), find a point $a \in (\alpha, c)$ with $f(a) < 0$. Treat separately the cases $\alpha \in \mathbb{R}$ and $\alpha = -\infty$.
(d) In the same way, find a point $b \in (c, \beta)$ with $f(b) > 0$.
(e) Explain why Theorem 9.1 (Bolzano's Theorem) can be applied to $f$ on $[a, b]$, and conclude that $f$ has a zero in $(\alpha, \beta)$.
(f) Prove that the zero is unique if $f$ is strictly monotone on $(\alpha, \beta)$.

(10) **Bolzano's Theorem at work.**
(a) Show that the equation $x^5 + x - 1 = 0$ has exactly one real solution, and that it lies in $(0, 1)$.
(b) Show that the equation $\cos x = x$ has exactly one solution in $[0, \frac{\pi}{2}]$.
(c) Show that every polynomial of odd degree has at least one real zero.
(d) *Bisection.* Perform three steps of the bisection method (as in the proof of Theorem 9.1) for $f(x) = x^3 - 2$ on $[1, 2]$, and give an interval of length $\frac{1}{8}$ containing $\sqrt[3]{2}$. How many steps guarantee that the midpoint of the last interval approximates $\sqrt[3]{2}$ with an error smaller than $10^{-3}$?
(e) The function $f(x) = \frac{1}{x}$ satisfies $f(-1) < 0 < f(1)$, but it has no zero in $[-1, 1]$. Which hypothesis of Bolzano's Theorem fails?
(f) *Kepler's equation.* The position of a planet or a satellite on its elliptic orbit at time $t$ is found by solving Kepler's equation

$$M = E - e \sin E$$

for the unknown $E \in \mathbb{R}$, where $M \in \mathbb{R}$ is proportional to $t$ and $e \in [0, 1)$ is the eccentricity of the orbit. Show that for every $M \in \mathbb{R}$ the equation has at least one solution $E \in \mathbb{R}$. *(It is in fact unique, but we will prove this later.)*

(11) **Proofs with the Intermediate Value Theorem and Weierstrass' Theorem.**
(a) The lecture notes prove the Intermediate Value Theorem (Theorem 9.4) only in the case $f(a) < f(b)$. Prove the case $f(a) > f(b)$.
(b) Let $f : [0, 1] \to [0, 1]$ be continuous. Prove that $f$ has a *fixed point*: there exists $x_0 \in [0, 1]$ such that $f(x_0) = x_0$. *Hint: consider* $g(x) = f(x) - x$.
(c) Let $f$ be continuous on $[a, b]$ with $f(x) > 0$ for all $x \in [a, b]$. Prove that there exists $m > 0$ such that $f(x) \geq m$ for all $x \in [a, b]$. Is the same true on $(0, 1]$ instead of $[a, b]$?
(d) Let $f : \mathbb{R} \to \mathbb{R}$ be continuous with $\lim_{x \to \pm\infty} f(x) = 0$ and $f(0) > 0$. Prove that $f$ attains a maximum on $\mathbb{R}$.
(e) Let $f$ be continuous on an interval $I$ and suppose that $f(x) \in \mathbb{Q}$ for all $x \in I$. Prove that $f$ is constant on $I$. *Hint: if $r < s$ are rational, then $r + \frac{s-r}{\sqrt{2}}$ is an irrational number between $r$ and $s$.*

(12) **The Intermediate Value Theorem in practice.**
(a) A hiker starts walking up a mountain path at 8:00 and reaches the summit at 17:00. The next day she starts walking down the same path at 8:00 and reaches the bottom at 17:00 (her speed may vary, and she may stop to rest). Prove that there is a point of the path that she passes at the same time of the day on both days. *Hint: Corollary 9.3.*
(b) The temperature along a circular ring (for example, along the equator) is a continuous function $T(\theta)$ of the angle $\theta$, with $T(\theta + 2\pi) = T(\theta)$. Prove that there are two diametrically opposite points of the ring with the same temperature. *Hint: consider* $g(\theta) = T(\theta) - T(\theta + \pi)$ *on* $[0, \pi]$.

## Homework 7 Solutions

(1) (a) Multiplying out and collecting equal powers of $x$,

$$p_n(x)\,q_m(x) = \sum_{k=0}^{n+m} c_k x^k, \qquad c_k = \sum_{\substack{i+j=k \\ 0\le i\le n,\ 0\le j\le m}} a_i b_j,$$

i.e. $c_k$ is the sum of all products $a_i b_j$ whose indices add up to $k$; for instance $c_0 = a_0 b_0$, $c_1 = a_0 b_1 + a_1 b_0$, $c_2 = a_0 b_2 + a_1 b_1 + a_2 b_0$ (where we set $a_i = 0$ for $i > n$ and $b_j = 0$ for $j > m$, since $n$ and $m$ may be different). The highest power is $x^{n+m}$, with coefficient $c_{n+m} = a_n b_m \neq 0$, so the product has **degree** $n + m$.

(b) All statements are as $x \to 0$. Multiplying out,

$$f\,g = p_n\,q_m + p_n\,o(x^m) + q_m\,o(x^n) + o(x^n)\,o(x^m).$$

A polynomial is bounded near 0 (on $(-1,1)$, $|p_n(x)| \le |a_0| + \cdots + |a_n|$), so by Proposition 8.1(5) $p_n\,o(x^m) = o(x^m)$ and $q_m\,o(x^n) = o(x^n)$. By Corollary 8.2, $o(x^n)\,o(x^m) = o(x^{n+m})$. Since $N \le n$, $N \le m$ and $N \le n + m$, each of these three functions is also $o(x^N)$ (a function that is $o(x^k)$ with $k \ge N$ is $o(x^N)$, as in the proof of Corollary 8.2), and their sum is $o(x^N)$ by Proposition 8.1(1). Finally,

$$p_n\,q_m = r + \sum_{k=N+1}^{n+m} c_k x^k,$$

and each $c_k x^k$ with $k > N$ is $o(x^N)$ (near 0 bigger powers are negligible). Hence

$$f(x)\,g(x) = r(x) + o(x^N), \qquad x \to 0.$$

(c) Here $n = 2$, $m = 1$, so $N = 1$. We have $(1 + 2x - x^2)(3 - x) = 3 + 5x - 5x^2 + x^3$, and keeping the terms of degree at most 1,

$$f(x)\,g(x) = \mathbf{3 + 5x + o(x)}, \qquad x \to 0.$$

The coefficient of $x^2$ **cannot be determined**: the remainder $o(x)$ of $g$ can be any function negligible with respect to $x$, for instance $cx^2$ for any $c \in \mathbb{R}$, and it contributes $1 \cdot cx^2$ to the product. So the information we have on $g$ is only precise up to order 1.

(2) Throughout we use the equivalences of Section 8.1.5 (as $y \to 0$: $\sin y \sim y$, $1 - \cos y \sim \frac{1}{2}y^2$, $\ln(1+y) \sim y$, $e^y - 1 \sim y$, $(1+y)^\alpha - 1 \sim \alpha y$). They remain true when $y$ is replaced by a function $u(x) \to 0$ with $u(x) \neq 0$ near the limit point (Theorem 6.12, case 2); for example $\sin(3x) \sim 3x$ as $x \to 0$. Then Proposition 8.3 allows us to replace factors by equivalent ones.

(a) $\sin(3x) \sim 3x$, $\ln(1 + 2x) \sim 2x$, $e^x - 1 \sim x$, so the limit equals $\lim_{x\to 0} \frac{3x \cdot 2x}{x \cdot x} = \mathbf{6}$.

(b) $1 - \cos(2x) \sim \frac{1}{2}(2x)^2 = 2x^2$ and $\ln(1 + 3x) \sim 3x$, so the limit equals $\lim_{x\to 0} \frac{2x^2}{3x^2} = \mathbf{\frac{2}{3}}$.

(c) $\sqrt[3]{1 + 6x} - 1 = (1 + 6x)^{1/3} - 1 \sim \frac{1}{3} \cdot 6x = 2x$ and $\sin(4x) \sim 4x$, so the limit equals $\frac{2}{4} = \mathbf{\frac{1}{2}}$.

(d) $e^{x^2} - 1 \sim x^2$ and $1 - \cos x \sim \frac{1}{2}x^2$, so the limit equals $\frac{1}{1/2} = \mathbf{2}$.

(e) In the numerator $\ln(1 + 3x) \sim 3x$ and $x^2 = o(x) = o(\ln(1 + 3x))$ (Corollary 8.5); in the denominator $\sin^2 x \sim x^2 = o(2x)$. By Corollary 8.4 the terms $x^2$ and $\sin^2 x$ can be dropped:

$$\lim_{x\to 0} \frac{\ln(1 + 3x) + x^2}{2x + \sin^2 x} = \lim_{x\to 0} \frac{\ln(1 + 3x)}{2x} = \lim_{x\to 0} \frac{3x}{2x} = \mathbf{\frac{3}{2}}.$$

(f) As $x \to +\infty$, $u = \frac{1}{x} \to 0^+$ and $u \neq 0$, so $1 - \cos\frac{1}{x} \sim \frac{1}{2x^2}$. Hence the limit equals $\lim_{x\to+\infty} x^2 \cdot \frac{1}{2x^2} = \mathbf{\frac{1}{2}}$.

(g) $\ln\left(1 + \frac{2}{x}\right) \sim \frac{2}{x}$ as $x \to +\infty$, so the limit equals $\lim x \cdot \frac{2}{x} = \mathbf{2}$.

(h) Write $\left(1 + \frac{3}{x}\right)^{2x} = e^{2x \ln(1+3/x)}$. As $x \to +\infty$, $\ln\left(1 + \frac{3}{x}\right) \sim \frac{3}{x}$, so by Proposition 8.3

$$\lim_{x\to+\infty} 2x \ln\left(1 + \frac{3}{x}\right) = \lim_{x\to+\infty} 2x \cdot \frac{3}{x} = 6.$$

Since $e^y$ is continuous at 6 (Theorem 6.12, case 1), the limit is $\boldsymbol{e^6}$.

(i) Let $y = \frac{\pi}{2} - x \to 0^+$. Then $\cos x = \sin y$ and $\tan x = \frac{\sin x}{\cos x} = \frac{\cos y}{\sin y}$, so

$$\frac{1}{\cos x} - \tan x = \frac{1 - \cos y}{\sin y} \sim \frac{\frac{1}{2}y^2}{y} = \frac{y}{2} \to \mathbf{0}.$$

(j) Let $a_n = \frac{1}{n} \to 0$, $a_n \neq 0$. Since $\lim_{y\to 0} \frac{e^y - 1}{y} = 1$, Theorem 6.15(7b) gives $n\left(e^{1/n} - 1\right) = \frac{e^{a_n} - 1}{a_n} \to \mathbf{1}$.

(k) With $a_n = \frac{\pi}{n} \to 0$, $a_n \neq 0$: $n^2\left(1 - \cos\frac{\pi}{n}\right) = \pi^2 \cdot \frac{1 - \cos a_n}{a_n^2} \to \pi^2 \cdot \frac{1}{2} = \boldsymbol{\frac{\pi^2}{2}}$ (Theorem 6.15(7b) with $g(y) = \frac{1-\cos y}{y^2}$, whose limit at 0 is $\frac{1}{2}$).

(3) (a) Proposition 8.3 allows us to replace a function by an equivalent one only in **products and quotients**, not in sums or differences (see the Warning in Section 8.1.5). The numerator $\sin(2x) - 2\sin x$ is a difference, and replacing each term by an equivalent one gives $2x - 2x = 0$, which is not equivalent to the numerator (no function is equivalent to 0): all the information about the size of the numerator is lost in the cancellation.

(b) Using $\sin(2x) = 2\sin x \cos x$, the numerator is a *product*:

$$\sin(2x) - 2\sin x = 2\sin x\,(\cos x - 1) \sim 2x \cdot \left(-\frac{x^2}{2}\right) = -x^3,$$

so the limit equals $\lim_{x\to 0} \frac{-x^3}{x^3} = \boldsymbol{-1}$.

(4) (a) We use (Section 8.2): $\log_a x = o(x^s)$ and $x^s = o(b^x)$ as $x \to +\infty$ ($a, b > 1$, $s > 0$), and $b^n = o(n!)$, $n! = o(n^n)$, $n^s = o(b^n)$ as $n \to \infty$.

(i) Dividing by $e^x$: $\frac{x^5/e^x + 1}{2 + x^2/e^x} \to \frac{0+1}{2+0} = \boldsymbol{\frac{1}{2}}$.

(ii) Dividing by $3^x$: $\frac{x^3/3^x + (2/3)^x}{1 + x^{10}/3^x} \to \frac{0+0}{1+0} = \mathbf{0}$, since $\left(\frac{2}{3}\right)^x \to 0$ (base smaller than 1).

(iii) $\ln x = o(x^{1/2})$, so the limit is $\mathbf{0}$.

(iv) $\ln x = o(x^2)$, so the limit is $\mathbf{0}$.

(v) We have $n! = o(n^n)$ and $3^n = o(n!)$, hence also $3^n = o(n^n)$, since $\frac{3^n}{n^n} = \frac{3^n}{n!} \cdot \frac{n!}{n^n} \to 0$; moreover $\frac{n^{10}}{n^n} = n^{10-n} \leq \frac{1}{n}$ for $n \geq 11$, so $n^{10} = o(n^n)$. Dividing by $n^n$: $\frac{n!/n^n + 3^n/n^n}{1 + n^{10}/n^n} \to \frac{0+0}{1+0}$, so the limit is $\mathbf{0}$.

(b) By Section 8.2, as $n \to \infty$ we have $\log_2 n = o(n^s)$ for every $s > 0$; with $s = 2$ this gives $n\log_2 n = n \cdot o(n^2) = o(n^3)$. Moreover $n^3 = o(2^n)$ and $2^n = o(n!)$. So, from the fastest to the slowest: $\boldsymbol{n\log_2 n}$, $\boldsymbol{n^3}$, $\boldsymbol{2^n}$, $\boldsymbol{n!}$. For $n = 50$ the numbers of operations and the times at $10^9$ operations per second are:

| | operations | time |
|---|---|---|
| $n\log_2 n$ | $\approx 282$ | $\approx 2.8 \cdot 10^{-7}$ s |
| $n^3$ | 125 000 | $1.25 \cdot 10^{-4}$ s |
| $2^n$ | $\approx 1.13 \cdot 10^{15}$ | $\approx 1.13 \cdot 10^6$ s $\approx$ 13 days |
| $n!$ | $\approx 3.0 \cdot 10^{64}$ | $\approx 3.0 \cdot 10^{55}$ s $\approx 10^{48}$ years |

The order "for large $n$" already decides which algorithm is usable in practice.

(5) (a) Rationalizing, $u(x) := \sqrt{x^2 - 1} - x = \frac{-1}{\sqrt{x^2-1} + x}$ for $x \geq 1$. Since $\frac{\sqrt{x^2-1}+x}{2x} = \frac{1}{2}\left(\sqrt{1 - \frac{1}{x^2}} + 1\right) \to 1$, we have $u(x) \sim -\frac{1}{2x}$; in particular $u(x) \to 0$ and $u(x) \neq 0$. Hence $\sin u(x) \sim u(x) \sim -\frac{1}{2x}$ as $x \to +\infty$ (Theorem 6.12, case 2, with $\frac{\sin y}{y} \to 1$). So $f$ is infinitesimal of **order** 1 with respect to $\frac{1}{x}$, with **principal part** $-\frac{1}{2x}$:

$$\sin\left(\sqrt{x^2 - 1} - x\right) = -\frac{1}{2x} + o\left(\frac{1}{x}\right), \qquad x \to +\infty.$$

(b) Since $2\ln 3 = \ln 9$, $f(x) = \ln\left(1 + \frac{1}{9}\sin\frac{2}{x}\right)$. As $x \to +\infty$, $u(x) = \frac{1}{9}\sin\frac{2}{x} \to 0$ and $u(x) \neq 0$ for $x > \frac{2}{\pi}$, so $f(x) \sim u(x)$; moreover $\sin\frac{2}{x} \sim \frac{2}{x}$. Hence $f(x) \sim \frac{2}{9x}$, i.e. $f(x) = \frac{2}{9x} + o(x^{-1})$: $b = \boldsymbol{\frac{2}{9}}$, $\alpha = 1$.

(c) $f(x) = \frac{e^x-1-x^2}{1+x^2}$. As $x \to 0$, $e^x - 1 \sim x$ and $x^2 = o(x)$, so the numerator is $\sim x$ (Corollary 8.4), while the denominator tends to 1. Hence $f(x) \sim x$: **order 1, principal part** $x$, i.e. $f(x) = x + o(x)$.

(6) (a) $f$ has *order* $\alpha > 0$ at $x_0$ with respect to $\varphi$ if $f \asymp \varphi^\alpha$ as $x \to x_0$, i.e.

$$\lim_{x\to x_0} \frac{f(x)}{|\varphi(x)|^\alpha} = \ell \in \mathbb{R} \setminus \{0\}$$

(with $|\varphi|$ in place of $\varphi$ when $\varphi$ may be negative and $\alpha$ is not an integer, as in Section 8.3).

(b) Suppose $\frac{f}{|\varphi|^\alpha} \to \ell_1 \neq 0$ and $\frac{f}{|\varphi|^\beta} \to \ell_2 \neq 0$ as $x \to x_0$, with $\alpha, \beta > 0$. By Theorem 6.2 (local sign), $\frac{f}{|\varphi|^\beta} \neq 0$ on a deleted neighborhood of $x_0$, so there we may write

$$|\varphi(x)|^{\beta-\alpha} = \frac{|\varphi(x)|^\beta}{|\varphi(x)|^\alpha} = \frac{f(x)/|\varphi(x)|^\alpha}{f(x)/|\varphi(x)|^\beta} \xrightarrow{x\to x_0} \frac{\ell_1}{\ell_2} \in \mathbb{R} \setminus \{0\}.$$

Suppose by contradiction that $\alpha \neq \beta$. If $f$ is infinitesimal, then $|\varphi(x)| \to 0^+$ ($\varphi$ is $x - x_0$ or $\frac{1}{x}$), so $|\varphi|^{\beta-\alpha} \to 0$ if $\beta > \alpha$ and $\to +\infty$ if $\beta < \alpha$. If $f$ is infinite, then $|\varphi(x)| \to +\infty$ and the two cases are swapped. In all cases the limit is 0 or $+\infty$, not a nonzero real number: a contradiction. Hence $\alpha = \beta$.

(7) (a) (i) $\sqrt{1+x^2} - 1 = (1+x^2)^{1/2} - 1 \sim \frac{1}{2}x^2$: **order 2, principal part** $\frac{1}{2}x^2$.

(ii) With $y = \sqrt{x} \to 0^+$: $1 - \cos y \sim \frac{1}{2}y^2 = \frac{x}{2}$: **order 1, principal part** $\frac{x}{2}$.

(iii) As $x \to 0^+$: $\frac{x^3}{\sqrt{x}} = x^{5/2} \to 0$, so $x^3 = o(\sqrt{x})$ and $x^3 + \sqrt{x} \sim \sqrt{x} = x^{1/2}$: infinitesimal of **order** $\frac{1}{2}$**, principal part** $\sqrt{x}$. As $x \to +\infty$: $\sqrt{x} = o(x^3)$, so $x^3 + \sqrt{x} \sim x^3$: infinite of **order 3, principal part** $x^3$.

(iv) $\frac{x^2+1}{x-1} \Big/ \frac{1}{x-1} = x^2 + 1 \to 2$, so $\frac{x^2+1}{x-1} \sim \frac{2}{x-1}$: infinite of **order 1, principal part** $\frac{2}{x-1}$.

(b) $d(h) = \sqrt{2Rh + h^2} = \sqrt{2R}\,\sqrt{h}\,\sqrt{1 + \frac{h}{2R}}$, and the last factor tends to 1 as $h \to 0^+$. Hence $d(h) \to 0$ and $d(h) \sim \sqrt{2R}\,h^{1/2}$: **order** $\frac{1}{2}$**, principal part** $\sqrt{2R}\,\sqrt{h}$. With $R = 6371$ km and $h$ in meters ($h/1000$ km),

$$d \approx \sqrt{2 \cdot 6371 \cdot \frac{h}{1000}}\ \text{km} = \sqrt{12.742\,h}\ \text{km} \approx 3.57\sqrt{h}\ \text{km}.$$

For $h = 1.7$ m: $d \approx \mathbf{4.7}$ km; for $h = 100$ m: $d \approx \mathbf{35.7}$ km. As $h \to +\infty$, $\frac{d(h)}{h} = \sqrt{1 + \frac{2R}{h}} \to 1$, so $d(h) \sim h$: $d$ is infinite of order 1 with respect to $h$ (from very far away, the distance to the horizon is about the distance to the Earth).

(8) (a) We use the Proposition of Section 8.4: $y = ax + b$ is an asymptote at $+\infty$ iff $a = \lim \frac{f(x)}{x} \in \mathbb{R}$ and $b = \lim(f(x) - ax) \in \mathbb{R}$ (similarly at $-\infty$).

(i) Domain $x \neq -2$. Dividing, $f(x) = x + 1 - \frac{3}{x+2}$. At $-2$ the numerator tends to $-3$, so $f \to -\infty$ as $x \to -2^+$ and $f \to +\infty$ as $x \to -2^-$: **vertical asymptote** $x = -2$. Since $f(x) - (x+1) = -\frac{3}{x+2} \to 0$: **oblique asymptote** $y = x + 1$ **at** $\pm\infty$.

(ii) Domain $x \leq -4$ or $x \geq 0$; $f$ is continuous there, with finite values at $-4$ and 0: no vertical asymptotes. At $+\infty$: $\frac{f(x)}{x} = \sqrt{1 + \frac{4}{x}} \to 1$ and $f(x) - x = \frac{4x}{\sqrt{x^2+4x}+x} = \frac{4}{\sqrt{1+4/x}+1} \to 2$: $y = x + 2$ **at** $+\infty$. At $-\infty$ (where $\sqrt{x^2} = -x$): $\frac{f(x)}{x} = -\sqrt{1 + \frac{4}{x}} \to -1$ and $f(x) + x = \frac{4x}{\sqrt{x^2+4x}-x} = \frac{4x}{-x\left(\sqrt{1+4/x}+1\right)} \to -2$: $y = -x - 2$ **at** $-\infty$.

(iii) Domain $x \neq 0$. As $x \to 0^-$, $\frac{1}{x} \to -\infty$, so $e^{1/x} \to 0$ and $f \to 0$. As $x \to 0^+$, with $y = \frac{1}{x} \to +\infty$, $f = \frac{e^y}{y} \to +\infty$ (since $y = o(e^y)$): **vertical asymptote** $x = 0$ (from the right only). At $\pm\infty$: $\frac{f(x)}{x} = e^{1/x} \to 1$ and $f(x) - x = x\left(e^{1/x} - 1\right) = \frac{e^y-1}{y}$ with $y = \frac{1}{x} \to 0$, $y \neq 0$, so it tends to 1: $y = x + 1$ **at** $\pm\infty$.

(iv) $f$ is continuous on $\mathbb{R}$ and even. At $+\infty$: $\frac{f(x)}{x} = \arctan x \to \frac{\pi}{2}$, and, using the hint (for $x > 0$, $t = \arctan x \in (0, \frac{\pi}{2})$ and $\tan(\frac{\pi}{2} - t) = \frac{1}{x}$),

$$f(x) - \frac{\pi}{2}x = -x\left(\frac{\pi}{2} - \arctan x\right) = -x \arctan\frac{1}{x} = -\frac{\arctan y}{y} \to -1,$$

where $y = \frac{1}{x} \to 0^+$ and $\frac{\arctan y}{y} = \frac{t}{\tan t} = \frac{t}{\sin t}\cos t \to 1$ (with $t = \arctan y \to 0$). So $y = \frac{\pi}{2}x - 1$ **at** $+\infty$ and, since $f$ is even, $y = -\frac{\pi}{2}x - 1$ **at** $-\infty$.

(v) Domain $[0, +\infty)$, $f$ continuous: no vertical asymptotes. As $x \to +\infty$, $f \to +\infty$ and, with $y = \sqrt{x}$, $\frac{f(x)}{x} = \frac{e^y}{y^2} \to +\infty$: **no asymptotes**.

(vi) $f$ is continuous on $\mathbb{R}$ and $f \to 0$ at $\pm\infty$: **horizontal asymptote** $y = 0$ **at** $\pm\infty$, and no others.

(vii) As $x \to 0^+$ and as $x \to \pi^-$, $\sin x \to 0^+$, so $\ln(\sin x) \to -\infty$: **vertical asymptotes** $x = 0$ **and** $x = \pi$. The domain is bounded, so there is nothing to check at $\pm\infty$.

(viii) $f$ is continuous on $\mathbb{R}$: no vertical asymptotes. At $\pm\infty$: $\frac{f(x)}{x} = \sqrt[3]{\frac{1}{x} - \frac{3}{x^2}} \to 0$, so $a = 0$; but $f(x) - 0 \cdot x = f(x) \to +\infty$, so $b$ is not finite: **no asymptotes**.

(ix) Domain $x \neq 1$. At $\pm\infty$: $\frac{x}{x-1} \to 1$, so $f \to \arctan 1 = \frac{\pi}{4}$: **horizontal asymptote** $y = \frac{\pi}{4}$ **at** $\pm\infty$. At 1: $\frac{x}{x-1} \to \pm\infty$ as $x \to 1^{\pm}$, so $f \to \pm\frac{\pi}{2}$; the one-sided limits are finite (a jump), so there is **no vertical asymptote**.

(b) (i) As $t \to +\infty$, $-\frac{t}{\tau} \to -\infty$, so $e^{-t/\tau} \to 0$ (Theorem 6.12, case 3) and $v(t) \to v_T(1-0) = \boldsymbol{v_T}$.

(ii) As $t \to +\infty$, $e^{-t/\tau} \to 0$, so

$$\frac{s(t)}{t} = v_T - \frac{v_T\tau\left(1 - e^{-t/\tau}\right)}{t} \to v_T, \qquad s(t) - v_T t = -v_T\tau\left(1 - e^{-t/\tau}\right) \to -v_T\tau.$$

Hence the oblique asymptote is $\boldsymbol{s = v_T(t - \tau)}$. After a long time the body moves (almost) with the terminal velocity $v_T$, but it is $v_T\tau$ **behind** a body that moved with velocity $v_T$ from the start, as if it had started $\tau$ later.

(9) (a) **Corollary 9.2.** Let $(\alpha, \beta) \subseteq \mathbb{R}$ be an interval ($\alpha$ may be $-\infty$ and $\beta$ may be $+\infty$), and let $f : (\alpha, \beta) \to \mathbb{R}$ be continuous with $\lim_{x\to\alpha^+} f(x) = \ell_\alpha$ and $\lim_{x\to\beta^-} f(x) = \ell_\beta$ (finite or infinite), where $\ell_\alpha < 0 < \ell_\beta$ or $\ell_\alpha > 0 > \ell_\beta$. Then $f$ has a zero in $(\alpha, \beta)$. Moreover, if $f$ is strictly monotone on $(\alpha, \beta)$, the zero is unique.

(b) Suppose $\ell_\alpha < 0 < \ell_\beta$; the case $\ell_\alpha > 0 > \ell_\beta$ is similar (or apply the first case to $-f$), and is omitted. Fix $c \in (\alpha, \beta)$.

(c) Since the right limit of $f$ at $\alpha$ is negative (a negative number or $-\infty$), by Theorem 6.2 (local sign, for one-sided limits) $f < 0$ on a right neighborhood of $\alpha$:

- if $\alpha \in \mathbb{R}$, there is $\delta > 0$ with $f(x) < 0$ for all $x \in (\alpha, \alpha + \delta) \cap (\alpha, \beta)$; take any $a$ with $\alpha < a < \min\{\alpha + \delta, c\}$;
- if $\alpha = -\infty$, there is $M$ with $f(x) < 0$ for all $x < M$, $x \in (\alpha, \beta)$; take any $a < \min\{M, c\}$.

In both cases $a \in (\alpha, c)$ and $f(a) < 0$.

(d) In the same way, $f > 0$ on a left neighborhood of $\beta$ (of the form $(\beta - \delta', \beta)$ if $\beta \in \mathbb{R}$, or $(M', +\infty)$ if $\beta = +\infty$), and we choose $b$ in it with $b > c$. Then $b \in (c, \beta)$ and $f(b) > 0$.

(e) We have $\alpha < a < c < b < \beta$, so $[a, b] \subseteq (\alpha, \beta)$ is a closed bounded interval on which $f$ is continuous, and $f(a) < 0 < f(b)$. By Theorem 9.1 there is $x_0 \in (a, b) \subseteq (\alpha, \beta)$ with $f(x_0) = 0$.

(f) If $f$ is strictly monotone on $(\alpha, \beta)$, it is injective there (Proposition 4.1), so it takes the value 0 at most once: the zero is unique.

(10) (a) $f(x) = x^5 + x - 1$ is continuous on $\mathbb{R}$ (a polynomial), with $f(0) = -1 < 0 < 1 = f(1)$; by Theorem 9.1 it has a zero in $(0, 1)$. Both $x^5$ (an odd power) and $x$ are strictly increasing on $\mathbb{R}$, so $f$ is strictly increasing (Lemma 4.2), hence injective (Proposition 4.1): it has **exactly one** zero on $\mathbb{R}$, and it lies in $(0, 1)$ (numerically $x_0 \approx 0.755$).

(b) Apply Corollary 9.3 to $f(x) = \cos x$ and $g(x) = x$ on $[0, \frac{\pi}{2}]$: $f(0) = 1 > 0 = g(0)$ and $f(\frac{\pi}{2}) = 0 < \frac{\pi}{2} = g(\frac{\pi}{2})$, so there is $x_0 \in (0, \frac{\pi}{2})$ with $\cos x_0 = x_0$. Since cos is strictly decreasing on $[0, \frac{\pi}{2}]$ and $g$ is strictly increasing, the solution is **unique** (Corollary 9.3, and the endpoints are not solutions); numerically $x_0 \approx 0.739$.

(c) Let $p(x) = a_d x^d + \cdots + a_0$ with $d$ odd and $a_d \neq 0$; replacing $p$ by $-p$ if necessary, $a_d > 0$. Since $p(x) = a_d x^d \left(1 + \frac{a_{d-1}}{a_d x} + \cdots + \frac{a_0}{a_d x^d}\right)$ and the bracket tends to 1, we get $p(x) \to +\infty$ as $x \to +\infty$ and, $d$ being odd, $p(x) \to -\infty$ as $x \to -\infty$. $p$ is continuous on $(-\infty, +\infty)$, so by Corollary 9.2 it has a zero.

(d) $f(1) = -1 < 0 < 6 = f(2)$. Step 0: $c_0 = 1.5$, $f(1.5) = 1.375 > 0$, new interval $[1, 1.5]$. Step 1: $c_1 = 1.25$, $f(1.25) \approx -0.047 < 0$, new interval $[1.25, 1.5]$. Step 2: $c_2 = 1.375$, $f(1.375) \approx 0.600 > 0$, new interval **$[1.25, 1.375]$**, of length $\frac{1}{8}$, which contains $\sqrt[3]{2} \approx 1.2599$. After $n$ steps the interval has length $\frac{1}{2^n}$ and its midpoint is within $\frac{1}{2^{n+1}}$ of $\sqrt[3]{2}$. We need $2^{n+1} > 1000$, i.e. $n + 1 \geq 10$: $n = 9$ **steps** suffice.

(e) $f(x) = \frac{1}{x}$ is not defined at $0 \in [-1, 1]$, so it is **not continuous on** $[-1, 1]$; Bolzano's Theorem does not apply.

(f) Let $F(E) = E - e \sin E - M$, continuous on $\mathbb{R}$. Since $|\sin E| \leq 1$ and $0 \leq e < 1$, we have $E - 1 - M \leq F(E) \leq E + 1 - M$. As $E \to +\infty$ the left side tends to $+\infty$, and as $E \to -\infty$ the right side tends to $-\infty$; by Theorem 6.11, $F(E) \to +\infty$ as $E \to +\infty$ and $F(E) \to -\infty$ as $E \to -\infty$. By Corollary 9.2 on $(-\infty, +\infty)$, $F$ has a zero: Kepler's equation has a solution $E$ for every $M$.

(11) (a) Let $f$ be continuous on $[a, b]$ with $f(a) > f(b)$, and let $y_0$ lie between $f(b)$ and $f(a)$. If $y_0 = f(a)$ or $y_0 = f(b)$ there is nothing to prove; so let $f(b) < y_0 < f(a)$. Let $g(x) = y_0$ for all $x$. Then $f(a) > g(a)$ and $f(b) < g(b)$, so by Corollary 9.3 there is $x_0 \in (a, b)$ with $f(x_0) = g(x_0) = y_0$. (Alternatively, apply the case proved in the notes to $-f$ and $-y_0$.)

(b) $g(x) = f(x) - x$ is continuous on $[0, 1]$, $g(0) = f(0) \geq 0$ and $g(1) = f(1) - 1 \leq 0$, since $0 \leq f \leq 1$. If $g(0) = 0$ take $x_0 = 0$; if $g(1) = 0$ take $x_0 = 1$. Otherwise $g(0) > 0 > g(1)$, and Theorem 9.1 gives $x_0 \in (0, 1)$ with $g(x_0) = 0$, i.e. $f(x_0) = x_0$.

(c) By Weierstrass' Theorem $f$ attains its minimum on $[a, b]$: there is $\xi \in [a, b]$ with $f(x) \geq f(\xi)$ for all $x \in [a, b]$. Take $m = f(\xi)$; then $m > 0$ by assumption. **On** $(0, 1]$ **this is false**: $f(x) = x$ is continuous and positive on $(0, 1]$, but $\inf_{(0,1]} f = 0$, so no $m > 0$ works (Weierstrass' Theorem needs a closed bounded interval).

(d) Since $f(0) > 0$ and $f \to 0$ at $+\infty$, there is $N_1$ with $|f(x)| < f(0)$ for all $x > N_1$; since $f \to 0$ at $-\infty$, there is $N_2$ with $|f(x)| < f(0)$ for all $x < N_2$. Let $M = \max\{N_1, -N_2, 1\} > 0$. By Weierstrass' Theorem, $f$ attains a maximum on $[-M, M]$ at some $\xi$; since $0 \in [-M, M]$, $f(\xi) \geq f(0)$. If $|x| > M$, then $x > N_1$ or $x < N_2$, so $f(x) \leq |f(x)| < f(0) \leq f(\xi)$. Hence $f(x) \leq f(\xi)$ for all $x \in \mathbb{R}$: $f(\xi)$ is the maximum of $f$ on $\mathbb{R}$.

(e) Suppose $f$ is not constant: there are $x_1, x_2 \in I$ with $r = f(x_1) < f(x_2) = s$, both rational. By Corollary 9.5, $f(I)$ is an interval, so $[r, s] \subseteq f(I)$. The number $z = r + \frac{s-r}{\sqrt{2}}$ lies in $(r, s)$ (since $0 < \frac{1}{\sqrt{2}} < 1$) and is irrational: otherwise $\sqrt{2} = \frac{s-r}{z-r}$ would be rational. So $z = f(x)$ for some $x \in I$, contradicting $f(x) \in \mathbb{Q}$. Hence $f$ is constant.

(12) (a) Let $L > 0$ be the length of the path, and let $u(t)$ and $d(t)$ be the distances (along the path) from the bottom at time $t \in [8, 17]$ on the first and on the second day. Both are continuous (the hiker does not jump), and

$$u(8) = 0 < L = d(8), \qquad u(17) = L > 0 = d(17).$$

By Corollary 9.3 there is $t_0 \in (8, 17)$ with $u(t_0) = d(t_0)$: at time $t_0$ she is at the same point of the path on both days. No assumption on her speed is needed.

(b) $g(\theta) = T(\theta) - T(\theta + \pi)$ is continuous on $[0, \pi]$, and, by periodicity,

$$g(\pi) = T(\pi) - T(2\pi) = T(\pi) - T(0) = -g(0).$$

If $g(0) = 0$, the points $\theta = 0$ and $\theta = \pi$ have the same temperature. Otherwise $g(0)$ and $g(\pi)$ have opposite signs, and Theorem 9.1 gives $\theta_0 \in (0, \pi)$ with $g(\theta_0) = 0$, i.e. $T(\theta_0) = T(\theta_0 + \pi)$: the diametrically opposite points $\theta_0$ and $\theta_0 + \pi$ have the same temperature.